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 Studierichting: Informatica
 Oefeningenreeks 3G nr. 2

$$\begin{aligned} f(x) &= (x-2)^3 + 1 \\ f'(x) &= 3(x-2)^2 \\ f''(x) &= 6(x-2) \end{aligned}$$

Nulpunten:

$$f(x) = 0 \Leftrightarrow (x-2)^3 + 1 = 0 \Leftrightarrow (x-2)^3 = -1 \Leftrightarrow x = 1$$

Extrema:

$$f'(x) = 0 \Leftrightarrow 3(x-2)^2 = 0 \Leftrightarrow x = 2$$

Buigpunten:

$$f''(x) = 0 \Leftrightarrow 6(x-2) = 0 \Leftrightarrow x = 2$$

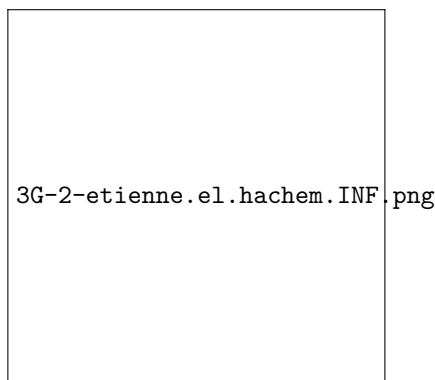
T.O.:

x	1		2	
$f(x)$	-	0	+	+
$f'(x)$	+	+	+	0 ⁽²⁾
$f''(x)$	-	-	-	0
$f(x)$	⌒	0	⌒	B
	↗	0	↗	(1)

Raaklijn in snijpunten:

In punt: $x = 1, y = 0$

Raaklijn: $y = 3x - 3$



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References